

Forward Backward Reasoning(For Mathematical Verifications)

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Definition

- **Forward Backward Reasoning** (FOBAR) enhances mathematical problem-solving with dual-directional logic.
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- FOBAR utilizes a blend of forward and backward reasoning, offering a comprehensive approach to tackling math problems

How Forward Backward Reasoning Works



Initiates with forward reasoning, generating multiple solutions

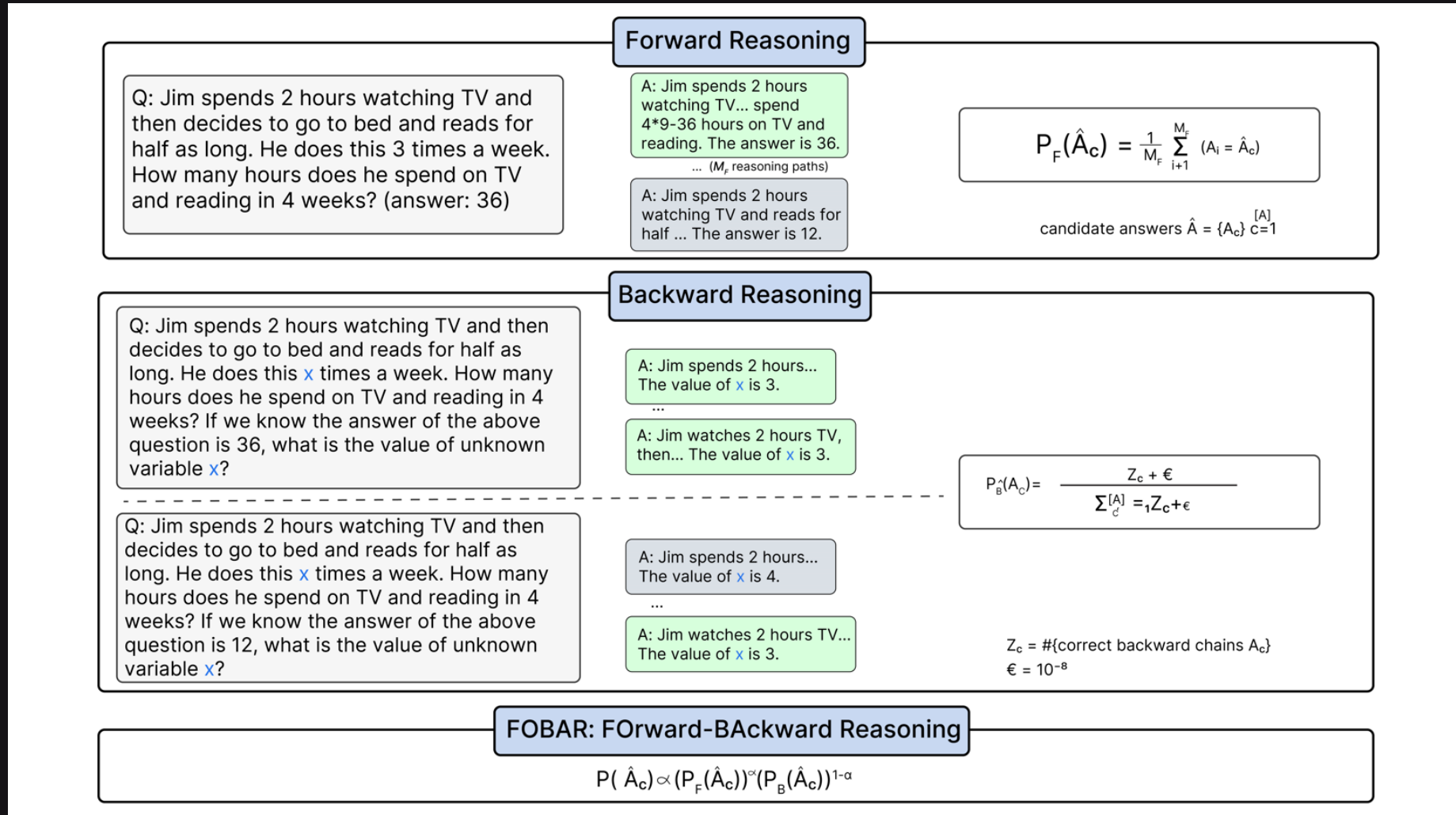


A possible solution is tested by tracing steps in reverse to confirm its viability.

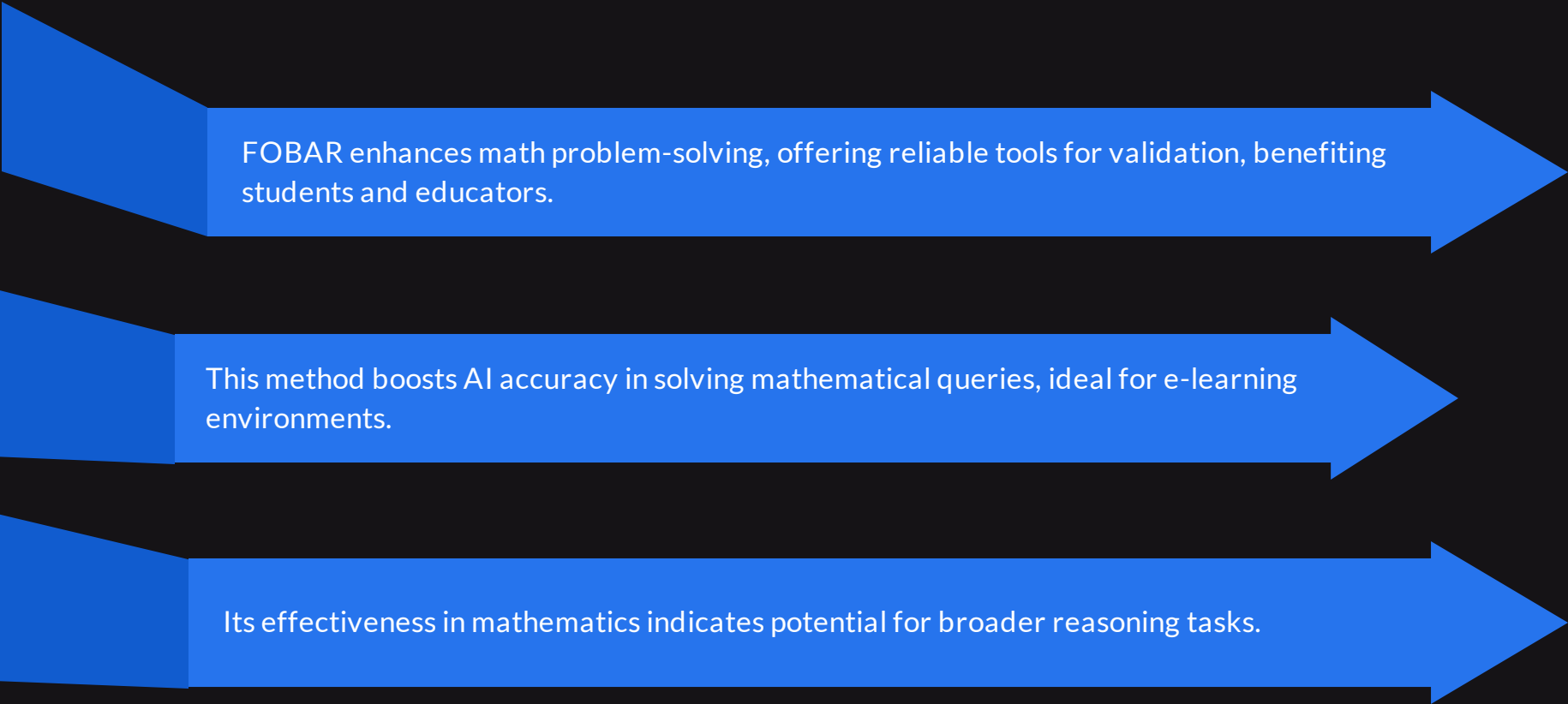


FOBAR consolidates both reasoning results, applying specialized calculations to determine the most probable correct answer.

Forward Backward Reasoning: Example



Forward Backward Reasoning: Practical Use-case



FOBAR enhances math problem-solving, offering reliable tools for validation, benefiting students and educators.

This method boosts AI accuracy in solving mathematical queries, ideal for e-learning environments.

Its effectiveness in mathematics indicates potential for broader reasoning tasks.

Thank You
